



Associations between Parenting and Cognitive and Language Abilities at 2 Years of Age Depend on Prenatal Exposure to Disadvantage

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Objective To investigate whether parenting or neonatal brain volumes mediate associations between prenatal social disadvantage (PSD) and cognitive/language abilities and whether these mechanisms vary by level of disadvantage.

Study design Pregnant women were recruited prospectively from obstetric clinics in St Louis, Missouri. PSD encompassed access to social (eg, education) and material (eg, income to needs, health insurance, area deprivation, and nutrition) resources during pregnancy. Neonates underwent brain magnetic resonance imaging. Mother-child dyads ($n = 202$) returned at age 1 year for parenting observations and at age 2 years for cognition/language assessments (Bayley Scales of Infant and Toddler Development, Third Edition). Generalized additive and mediation models tested hypotheses.

Results Greater PSD associated nonlinearly with poorer cognitive/language scores. Associations between parenting and cognition/language were moderated by disadvantage, such that supportive and nonsupportive parenting behaviors related only to cognition/language in children with lesser PSD. Parenting mediation effects differed by level of disadvantage: both supportive and nonsupportive parenting mediated PSD-cognition/language associations in children with lesser disadvantage, but not in children with greater disadvantage. PSD-associated reductions in neonatal subcortical grey matter ($\beta = 0.19$; $q = 0.03$), white matter ($\beta = 0.23$; $q = 0.02$), and total brain volume ($\beta = 0.18$; $q = 0.03$) were associated with lower cognition, but did not mediate the associations between PSD and cognition.

Conclusions Parenting moderates and mediates associations between PSD and early cognition and language, but only in families with less social disadvantage. These findings, although correlational, suggest that there may be a critical threshold of disadvantage, below which mediating or moderating factors become less effective, highlighting the importance of reducing disadvantage as primary prevention. (*J Pediatr* 2025;276:114289).

Social disadvantage, or a lack of access to material and social resources, is associated consistently with variability in cognitive and language abilities¹⁻³ that emerges early in development.^{4,5} Recently, prenatal social disadvantage (PSD) has been associated negatively with early cognitive⁶ and language⁷ abilities. These early disparities each contribute to a socioeconomic achievement gap⁸ that increases with age.⁹ To identify effective resilience and intervention targets, it is important to better understand the mechanisms through which PSD is associated with early cognitive and language development.

One mechanism might be via the impact of PSD on fetal brain development. PSD has been related to decreased neonatal gray, white, and total brain volumes.¹⁰ In older children, such decreases (eg, frontal and temporal lobes) mediate associations between postnatal social disadvantage and cognition.¹¹ These neural phenotypes at birth may also mediate associations between PSD and cognition. Similarly, neonatal brain volumes¹² have been associated with early language abilities; however, mediations have not been examined.

Another mechanism may be via parenting behaviors, which are influenced by correlates of social disadvantage.¹³⁻¹⁵ Parenting behaviors (eg, maternal sensitivity) are associated both with cognitive¹⁶ and language development.^{17,18} Moreover, responsive parenting behaviors have mediated the relations of postnatal social

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EDF	Effective degree of freedom
GAM	Generalized additive model
GM	Gray matter
RLRT	Restricted likelihood ratio test
SES	Socioeconomic status
TOPF	Test of Premorbid Function

disadvantage to cognitive abilities (eg, cognitive flexibility, inhibition¹⁹) in older children. Although parental mediations of social disadvantage and early cognitive²⁰ and language¹⁸ development seem promising, existing investigations have been cross-sectional with contemporaneous measurements of parenting with social disadvantage or with outcomes.^{18,20} Thus, longitudinal studies are needed to test whether parenting behaviors may mediate PSD similarly to early cognitive and language outcomes. Parenting is particularly influential in early life,^{21,22} and constitutes a potentially malleable intervention target, and thus is an important mechanism to examine.

Finally, a complete mechanistic understanding requires specificity. Conceivably, mechanisms of PSD and cognition/language associations may vary by level of disadvantage, as do several existing disadvantage-outcome associations^{6,23,24} and their mediating pathways.^{25,26} This mechanistic specificity (eg, moderation) can inform the development of effective tailored preventative efforts.

In this prospective, longitudinal study, we investigated the mediating and moderating pathways between PSD and offspring cognitive and language outcomes through age 2 years. Given limited economic and social mobility in the US,²⁷ it is difficult to isolate associations specific to prenatal or postnatal social disadvantage, especially for postnatal outcomes. Nonetheless, PSD enables investigations of social disadvantage very early in development. We tested the following hypotheses: associations between PSD and cognition and language would be mediated by neonatal brain volume (of six a priori regions associated with PSD in this sample¹⁰) and parenting behaviors, and that the relationships between parenting or brain volumes and cognition/language would vary across levels of disadvantage (eg, disadvantage is a moderator).

Methods

Data were collected as part of the Early Life Adversity Biological Embedding (eLABLE) study.^{28,29} The study enrolled pregnant women and their 398 singleton offspring. Study exclusion criteria were multiple gestation, alcohol or substance use during pregnancy (except marijuana or tobacco), infections known to cause fetal congenital disease, and nonfluency in English. Women were oversampled from clinics serving low-income populations to enrich the sample for exposure to poverty. Women were enrolled early in pregnancy and completed assessments in each trimester. Infants were born between September 2017 and April 2020 and underwent brain magnetic resonance imaging scans shortly after birth. Mother-child dyads returned for follow-up assessments at ages 1 and 2 years. Written informed consent was obtained from mothers after describing the study protocols and before participation. Current analyses included all participants with complete cognitive and language assessments at year 2. All protocols were approved by the Institutional Review Board at Washington University in St. Louis.

Measures

Prenatal Social Disadvantage. As described in Luby et al,²⁸ PSD is a previously validated latent construct that includes variables relating to material and social resources during pregnancy: income-to-needs ratio, Area Deprivation Index, health insurance status, and maternal education and nutrition. Higher PSD scores represent greater disadvantage. See the Appendix for more details.

Bayley Scales of Infant and Toddler Development, Third Edition. The Bayley Scales of Infant and Toddler Development, Third Edition (Bayley-III), assesses cognitive, language, and motor development for children aged 1-42 months with good reliability and validity.³⁰ The primary dependent measures were age-normed composite scores for cognition, language, and motor abilities at the year 2 follow-up, wherein higher scores indicate more advanced development. To explore specificity to cognitive and language associations, motor abilities (which are unrelated to PSD in prior work³¹) were included as a control outcome.

Parenting Behaviors. Parenting behaviors were scored from three videotaped parent-child interaction tasks (adapted version of the Parent-Child Interaction Rating Scales).³² A supportive parenting behaviors composite ($\alpha = 0.861$) averaged scores for sensitivity and positive regard. A nonsupportive parenting behaviors composite ($\alpha = 0.726$) averaged scores for intrusiveness, detachment, and negative regard. Higher scores represent greater intensity and frequency of that behavior. Analyses used parenting behaviors at year 1 to enable mediation analyses with temporal precedence of parenting relative to cognitive and language outcomes (year 2). See the Appendix for more details and reliabilities.

Test of Premorbid Function. The Test of Premorbid Function (TOPF)³³ was used to assess maternal cognitive ability. It is a standardized task in which participants pronounce ≤ 70 words with irregular grapheme to phoneme translations. Age-normed TOPF standard scores are reported.

Magnetic Resonance Imaging. Magnetic resonance imaging data collection and processing were conducted as previously described¹⁰ and are further detailed in the Appendix. Briefly, T2-weighted images were obtained for infants during nonseparated sleep in the first weeks of life (mean, 3.2 weeks; social disadvantage = 1.9, range, 0-10 weeks). Prior work in this sample¹⁰ found associations between PSD and volumetric reductions in 6 neonatal brain metrics: total brain, cortical gray matter (GM), subcortical GM, total white matter, and cerebellar volumes and gyrification index.¹⁰ These 6 brain metrics were the a priori measures of interest.

Statistical Analysis

Covariates. Gestational age, infant sex assigned at birth, and age at follow-up were related to Bayley-III scores and were therefore included as covariates in all analyses. Postmenstrual

age at scan related to brain metrics and was a covariate in all neural models.

Modeling PSD and Child Outcomes. There is increasing evidence of nonlinear associations between socioeconomic status (SES) and brain development.^{11,34,35} Thus, predictors and covariates were added to generalized additive models (GAMs), nonparametric regressions wherein the relationship between the predictor and outcome is not assumed to be linear, allowing either nonlinear or linear relations to be detected and modeled in a data-driven fashion. Instead of beta coefficients, GAMs return an effective degree of freedom (EDF), which specifies the complexity of the relationship between predictor and outcome variables. An EDF of 1 is linear and higher EDFs indicate more nonlinearity. Effect sizes for nonlinear models were calculated using change in adjusted R^2 values ($\Delta\text{Adj } R^2$; the difference between the adjusted R^2 values of a model with and without the predictor[s] of interest), whereby positive values indicate that the predictor improves the model fit. The restricted likelihood ratio test (RLRT) (*RLRsim* package^{36,37}) determined whether the nonlinear fit was significantly better than the linear fit. If so (RLRT $P < .05$), GAMs were used in all subsequent analyses. If the nonlinear model was not a better fit (RLRT $P > .05$), linear regressions were used (*stats* package³⁸).

Neural Mediations. Regressions tested whether (1) each of the six a priori brain metrics related to outcomes, and (2) related brain metrics mediated and/or moderated PSD and outcome associations.

Parenting Behavior Mediation. Regressions tested whether (1) PSD related to parenting, (2) parenting related to outcomes, and (3) related parenting behaviors mediated and/or moderated PSD-outcome associations. Analyses were separate for supportive and nonsupportive parenting behaviors.

Disadvantage Moderations. Moderation was tested by adding an interaction between PSD and either parenting or neonatal brain metrics to each outcome. Because nonlinear moderations preclude follow-up with simple slopes, significant interactions were probed by examining associations between one predictor and the outcome separately in groups of lesser vs greater disadvantage, as defined by a median split. For significant mediators of PSD-cognition and language associations, we tested whether the mediation was significant in conditions of lesser vs greater disadvantage, as defined by a median split.

Mediations were conducted with GAM (nonlinear) or 'lm' (linear) objects in R (*mediation* package v4.5.0). Analyses used complete cases (listwise deletion). Results for all primary analyses were corrected using the False Discovery Rate corrections calculator.³⁹ See the Appendix for further details.

Results

At the 2-year follow up, Bayley-III scores were obtained for 202 children (Figure 1). Sample characteristics are detailed

in Table I. Race and ethnicity were identified from maternal race and ethnicity on the birth certificate. They are included in Table I for comparison to other samples in the literature; however, they were not included in any analyses. Children with and without Bayley-III data did not differ on any variables used in analyses. See the Appendix for details.

PSD and Primary Outcomes (n = 202)

PSD was nonlinearly related to cognition ($\Delta\text{Adj. } R^2 = 12.50\%$; EDF = 2.63; RLRT; $P = .03$; $q < .0001$) and language ($\Delta\text{Adj. } R^2 = 17.20\%$; EDF = 2.72; RLRT $P = .008$; $q < .0001$). As shown in Figure 2, greater PSD was associated with poorer cognition and language scores up to a certain level of PSD, beyond which additional disadvantage was not associated with further decreases in scores. PSD was not related to motor abilities ($\beta = -0.062$; $q = 0.37$). See Supplement Figure 1 (online; available at www.jpeds.com) for details about periods of significant change (eg, GAM derivatives).

Neural Mechanisms (n = 186)

Reduced subcortical GM, white matter, and total brain volumes at birth were linearly (all RLRT $P > .21$) associated with lower cognitive scores. No brain metric at birth related to language scores (Supplement Table 1, online; available at www.jpeds.com). None of the related brain metrics mediated associations between PSD and cognition or language (Supplement Table 2, online; available at www.jpeds.com).

Parenting Mechanisms (n = 135)

Greater PSD was linearly (RLRT $P = 1$) associated with fewer supportive parenting behaviors ($\beta = -0.62$; $q < 0.0001$) and more nonsupportive parenting behaviors ($\beta = 0.56$; $q < 0.0001$) at year 1 (Figure 3). Higher levels of supportive parenting behaviors were nonlinearly associated with higher cognition ($\Delta\text{Adj. } R^2 = 14.60\%$; EDF = 2.34; RLRT $P = .03$) and language scores ($\Delta\text{Adj. } R^2 = 18.90\%$; EDF = 2.35; RLRT $P = .02$), such that the relationship of supportive parenting to outcomes was stronger among higher supportive parenting scores (Figure 3). Higher levels of nonsupportive parenting behaviors were nonlinearly associated with poorer cognition ($\Delta\text{Adj. } R^2 = 11.11\%$; EDF = 4.17; $P = .03$) and language scores ($\Delta\text{Adj. } R^2 = 12.20\%$; EDF = 3.03; $P = .01$), such that the relation of nonsupportive parenting to cognition and language was strongest at lower levels of nonsupportive parenting (Figure 3). Further, lower supportive parenting mediated the relation of greater PSD to lower cognition and language scores (Figure 4), although nonsupportive parenting did not (Supplement Table 2, online; available at www.jpeds.com).

Disadvantage Moderations (n = 135)

PSD moderated relations of parenting to cognition (supportive interaction: $\Delta\text{Adj. } R^2 = 4.3\%$; EDF = 3.00; $q < 0.0001$; nonsupportive interaction: $\Delta\text{Adj. } R^2 = 4.4\%$; EDF = 5.40;

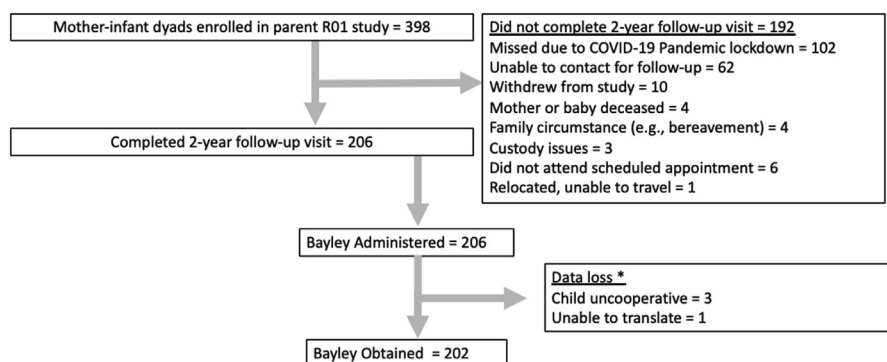


Figure 1. Flow chart of study participants. Flow chart of all participants that were initially enrolled in the Early Life Adversity Biological Embedding (eLABE) study (n = 398) through participants who enrolled at wave 2 and are included in this study.

$q < 0.0001$) and to language (supportive interaction: $\Delta\text{Adj. } R^2 = 1.7\%$; $\text{EDF} = 3.00$; $q < 0.0001$; nonsupportive interaction: $\Delta\text{Adj. } R^2 = 2.0\%$; $\text{EDF} = 3.16$; $q < 0.0001$) scores (Figure 5). Specifically, follow-up analyses revealed that for children with lesser levels of disadvantage, supportive and nonsupportive parenting behaviors were respectively positively and negatively associated with cognition (supportive: $P = .0003$; nonsupportive: $P = .005$) and language (supportive: $P = .00007$; nonsupportive: $P = .007$). However, at greater levels of PSD, the association between parenting and cognition or language was no longer significant (all P s $> .05$). Disadvantage did not moderate relations of neonatal brain volumes to cognition or to language (all P s $> .05$). See the Appendix for detailed results.

Given the parenting moderation and mediation identified in this work, exploratory mediation analyses of parenting behaviors were examined separately within groups of lesser vs greater levels of disadvantage, defined by a median split of greater (n = 71) and lesser (n = 64) disadvantage. Parenting (both supportive and nonsupportive) mediated the relationship of PSD to cognitive and language outcomes in the group with lesser disadvantage, but not in the group with greater disadvantage (Table II).

Sensitivity Analyses

We tested whether the PSD-cognition and PSD-language associations were significant over and above the influence of maternal cognitive ability by adding TOPF to the model. PSD continued to predict cognition ($\Delta\text{Adj. } R^2 = 1.9\%$; $P = .036$; n = 152) and language ($\Delta\text{Adj. } R^2 = 3.6\%$; $P = .017$; n = 152) even after accounting for maternal cognitive ability (which was no longer a significant predictor in either model [cognition: $P = .21$; language: $P = .23$]). We did not include maternal cognitive ability in all models to avoid multicollinearity, because maternal cognitive ability was collinear with disadvantage ($r = -0.80$; $P < .00001$). Because the study sample included preterm children, we re-ran all main analyses excluding children born preterm. None of the results changed (see the Appendix, online; available at www.jpeds.com).

Discussion

PSD was nonlinearly associated with cognitive and language outcomes at age 2, such that increasing PSD negatively predicted outcomes to a point, beyond which additional PSD was not associated with further decrements. Disadvantage moderated the relationships between parenting behaviors and cognition and language, such that parenting behaviors were only related to cognition and language for children with lower levels of disadvantage. Supportive parenting behaviors at age 1 mediated the relationships of PSD to cognition and to language. In contrast, brain metrics from birth did not mediate PSD and cognition or language associations. The mediating role of parenting in the association varied by level of disadvantage, which suggests that a basic level of resources may be necessary for parenting interventions to affect these key developmental outcomes. This finding has implications for parenting interventions that are often applied universally, without consideration of psychosocial context. This finding should inform multistep prevention and intervention efforts in early childhood for the greatest efficacy.

A key finding of the study was that PSD-related outcomes were detectable in toddlerhood. Although postnatal disadvantage has been associated reliably with cognitive and language abilities, investigations of PSD associations with these abilities are sparse. Our finding that PSD was associated with lower language abilities is consistent with existing findings of poorer language development in toddlers as a function of PSD.^{7,40} We cannot rule out the contribution of postnatal social disadvantage to either association, as prenatal and postnatal social disadvantage were very stable in our sample and highly correlated ($r = 0.92$). Nonetheless, our findings add to the growing literature documenting that the effects of social disadvantage begin very early in life,⁴¹ and highlight developmental consequences of PSD that are discernable as early as 2 years of age. Our nonlinear findings also show that, past a certain threshold of PSD, we do not detect significant associations of PSD with cognition and language. This nonlinearity may have important implications

Table 1. Sample characteristics

Maternal characteristics		
Prenatal social disadvantage latent factor score (n = 202)	−0.06 (0.97)	[−2.22 to 1.55]
Income-to-needs ratio	2.94 (3.1)	[0.32 to 12.15]
National area deprivation index	69.78 (24.05)	[7 to 100]
Healthy Eating Index	58.47 (10.47)	[31.70 to 80.67]
Highest level of education	-	
<12th grade	7 (3.46)	
High school degree/GED	44 (21.8)	
Some college/vocational school	70 (34.6)	
College degree (4 years)	25 (12.4)	
Graduate degree	56 (27.7)	
Health insurance status	-	
Medicaid or Medicare	69 (34.1)	
Individual or group health insurance	113 (60.0)	
VA/military	1 (0.5)	
Uninsured	19 (9.4)	
TOPF (n = 152)	94.22 (16.24)	[68 to 124]
Supportive parenting behaviors at year 1 (n = 135)	4.76 (0.89)	[2.33 to 6.33]
Sensitivity	4.81 (0.95)	[2.33 to 6.67]
Positive regard	4.71 (0.97)	[2.00 to 7.00]
Nonsupportive parenting behaviors at year 1 (n = 135)	1.64 (0.50)	[1.00 to 3.67]
Negative regard	1.33 (0.52)	[1.00 to 3.67]
Intrusiveness	2.16 (0.89)	[1.00 to 5.00]
Detachment	1.41 (0.67)	[1.00 to 4.33]
Maternal race	-	
African American	120 (59.4)	
Chinese	77 (38.1)	
Korean	1 (0.5)	
Multiracial	1 (0.5)	
Other	1 (0.5)	
Pacific Islander	1 (0.5)	
White	77 (38.1)	
Maternal ethnicity	-	
Hispanic/Latina	5 (2.5)	
Non-Hispanic/non-Latina	197 (97.5)	
Child characteristics (n = 202)		
Gestational age (weeks)	37.89 (1.9)	[28 to 41]
Age at year 2 (years)	2.10 (0.14)	[1.91 to 2.64]
Sex assigned at birth	-	
Male	110 (54.5)	
Female	92 (45.5)	
Bayley-III Cognition scores at year 2	88.64 (12.74)	[55 to 125]
Bayley-III Language scores at year 2	89.05 (16.69)	[50 to 138]
Expressive language	8.74 (2.80)	[2 to 19]
Receptive language	7.48 (3.22)	[1 to 16]
Postmenstrual age at scan, weeks (n = 186)	41.11 (1.48)	[37 to 45]
Neonatal brain volumes, mm³ (n = 186)		
Total cortical GM	120 303.1 (15 939.3)	[7318.5 to 161 621.8]
Total cerebral white matter	185 282.3 (19 424.6)	[136 548.9 to 244 443.3]
Total cerebellum	28 273.7 (4507.9)	[15 645.7 to 45 875.2]
Subcortical GM	27 115.2 (2703)	[20 404.8 to 35 261]
Total brain volume	359 364.2 (39 404.7)	[263 109.5 to 472 884.4]
Mean gyrification index	1.97 (.10)	[1.63 to 2.20]

Values are mean ± SD [range] or number (%).

for understanding and addressing the effect of disadvantage on development, because it reveals that the effect of disadvantage varies by level. For example, interventions may fail to show improvements in cognitive and language develop-

ment if they do not first decrease disadvantage to a level where enhancing parenting behaviors could positively impact cognition and language in children, a consideration that is currently overlooked in most intervention programs.

This study demonstrated that supportive parenting behaviors mediated the relation of PSD with 2-year cognition and language development, although parenting had a stronger effect at lower levels of disadvantage. There are several investigations of parenting behaviors (eg, sensitivity, responsivity) mediating effect of SES to a variety of offspring outcomes (eg, executive function,^{19,42} academic achievement,⁴³ behavioral⁴³ and externalizing problems²⁵) in older children (eg, kindergarten,⁴³ school age,^{19,25} and adolescents⁴²). However, investigations in early life are sparse. With regard to cognition, one study²⁰ reported that maternal sensitivity (averaged across 1 and 2 years) mediated the relation of social disadvantage (maternal education, income-to-needs ratio, welfare status from ages 1 to 2 years) to child Bayley-III cognition scores at 3 years of age. With regard to language, Raviv et al¹⁸ reported that maternal sensitivity at 3 years of age mediated the relationship of early social disadvantage (maternal education and income-to-needs ratio at 1 month of age) to receptive and expressive language at 3 years of age. However, these studies used contemporaneous assessments (disadvantage and parenting²⁰ or parenting and child outcomes¹⁸), limiting directional inferences. Our results are consistent with these prior findings in early life and provide additional evidence for parenting mediating disadvantage-cognition and language associations by examining these associations in a sample with temporal precedence. Although these findings may suggest initially that parenting behaviors in the first year of life may be a key target for intervention, we must also consider the finding that disadvantage moderated the relationship between parenting and neurocognitive outcomes.

Disadvantage moderated the relationship of parenting to cognition and language, such that exposure to more supportive and fewer nonsupportive parenting behaviors were related to better cognitive and language outcomes for children with lesser levels of disadvantage. However, parenting was not related to cognition or language for children with greater levels of disadvantage. Further, our analyses revealed that both supportive and nonsupportive parenting behaviors mediated the relation of PSD to cognition and language in the group with lesser disadvantage, but parenting behaviors did not mediate these relations in the group with greater disadvantage. To our knowledge, this has not been described previously, but these findings are consistent with evidence that, at varying levels of SES, different mechanistic pathways influence mental health severity.²⁵ Essex et al²⁵ reported differential mediational pathways of SES to third grade externalizing symptoms in high SES (eg, parental histories of depression and psychopathology) and low/middle SES (eg, maternal distress during infancy). Our results suggest that, although enhancing parenting practices (through socioculturally informed^{15,44,45} and child specific⁴⁶⁻⁴⁸ practices) may be helpful for children from advantaged backgrounds, it may be less effective when caregivers and children

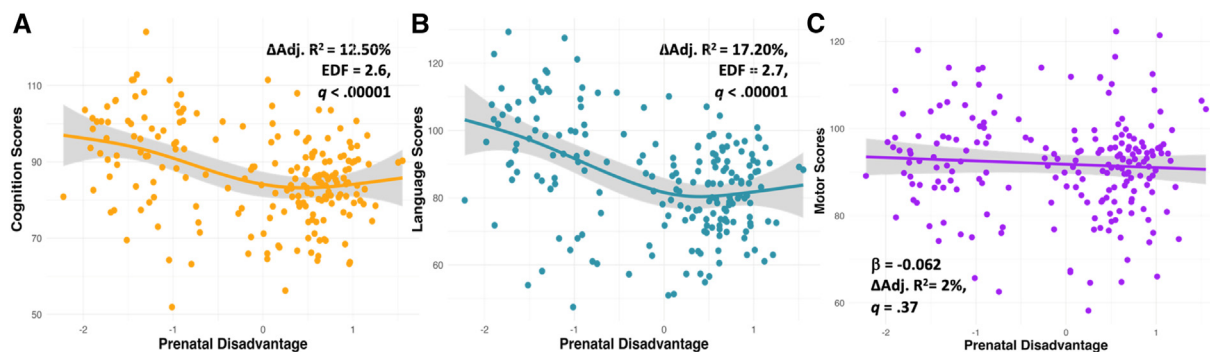


Figure 2. Scatterplots of associations between prenatal social disadvantage and (A) cognition, (B) language, and (C) motor outcomes at year 2. *EDF*, effective degrees of freedom. $\Delta \text{Adj. } R^2$ = the difference between the adjusted R^2 values of a model with and without the predictor of interest, where positive values indicate that the predictor of interest improves the model fit. See **Supplement Figure 1** (online; available at www.jpeds.com) for GAM derivatives.

experience the greatest levels of disadvantage. These effects need to be confirmed by causal study designs in the future, because they could underscore the need to provide families with a basic level of resources, because a minimum level of resources may be necessary for additional parenting interventions to improve child outcomes.

Although we found that parenting has weaker associations with child outcomes in settings of greater disadvantage, some studies report stronger associations between parenting be-

haviors and child outcomes (eg, executive function,⁴⁹ conduct problems⁵⁰) in highly disadvantaged settings. This result may be attributable to the fact that our sample is characterized by greater levels of disadvantage than previous studies (eg, where the mean income of the sample matches the mean income for the region).^{49,50} In such settings of great disadvantage, the cumulative effects of adverse daily environmental stressors may overwhelm the impact of parenting on child outcomes.^{50,51} This finding is consistent with other

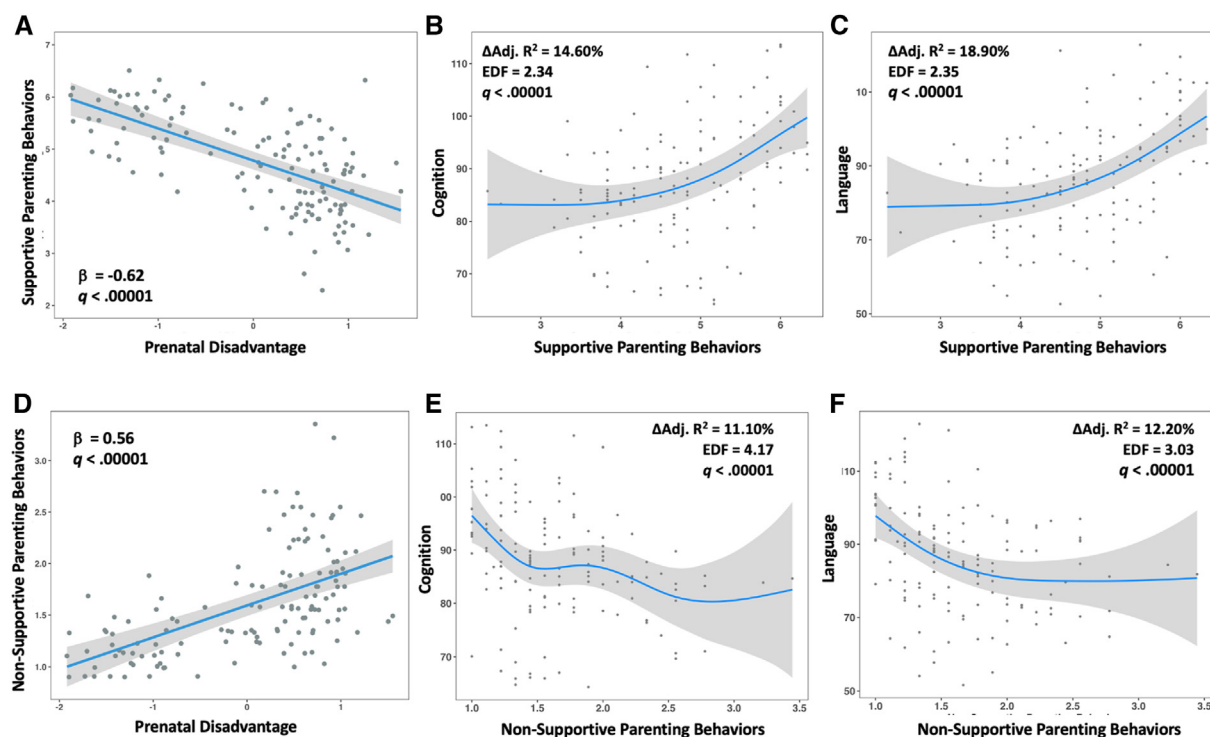


Figure 3. Associations of parenting behaviors with prenatal social disadvantage (A and D), cognition (B and E), and language (C and F). All figures illustrate the partial effect of independent variable (eg, controlling for covariates sex, age, and gestational age) on the dependent variable. *EDF*, expected degree of freedom for the shape of the smooth. Change in adjusted R^2 ($\Delta \text{Adj. } R^2$) = the difference between the adjusted R^2 values of a model with and without the predictor of interest, where positive values indicate that the predictor of interest improves the model fit.

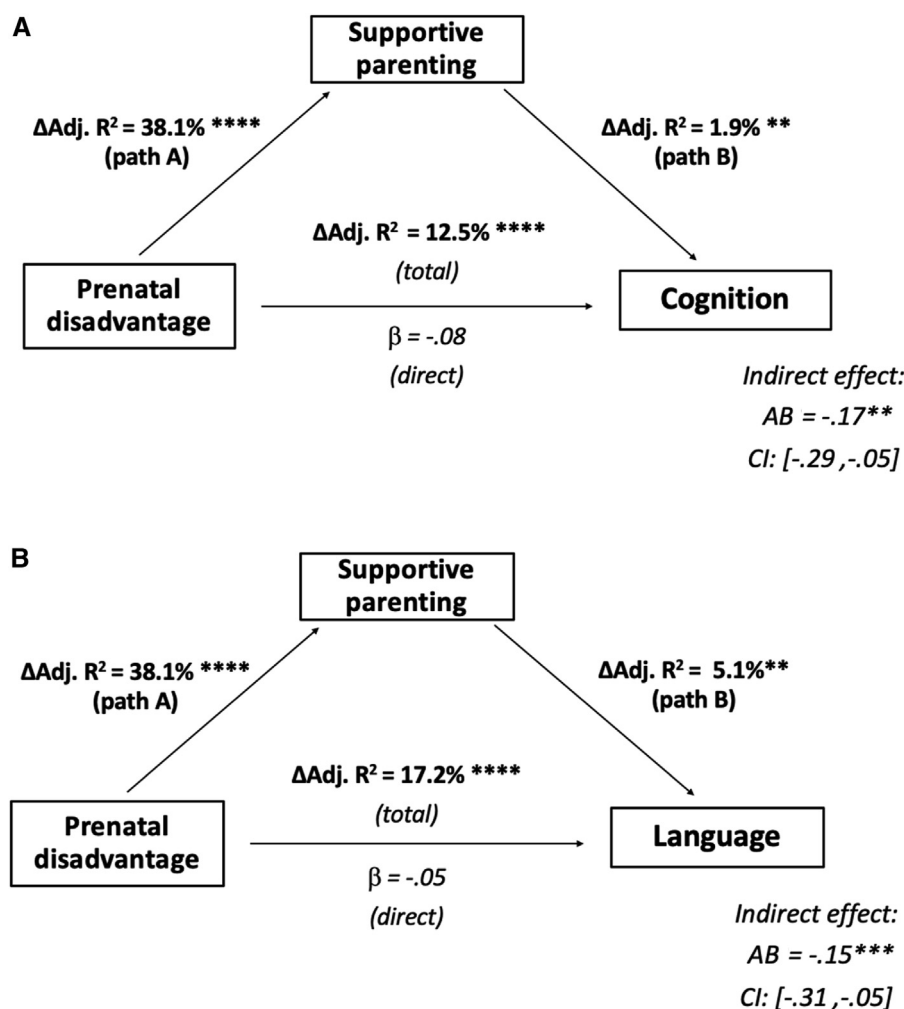


Figure 4. Supportive parenting behaviors mediate the relations of prenatal disadvantage to (A) cognition and (B) language. Change in adjusted R^2 values (Δ Adj. R^2) and significance levels for paths modeled with GAMs (RLRT $P < .05$). Standardized effect size and significance levels for direct and indirect effects. Confidence intervals for indirect effects. Path A describes the effect of the independent variable (prenatal disadvantage) to the mediator variable. Path B describes the unique effect of the mediator variable on the dependent variable. Direct effect indicates the unique effect of the independent variable (prenatal disadvantage) on the dependent variable, although controlling for the mediating variable. The indirect effect is the mediation effect. Note: GAMs do not return beta coefficients. Instead of a single beta coefficient that quantifies the fixed strength and direction of the linear relationship between x and y , GAMs allow the strength and direction of the relationship between x and y to vary across the range of x . Because of this, for paths modeled with GAMs, we instead provide the Δ Adj. R^2 as a measure of the effect size, which is the difference between the adjusted R^2 values of a model with and without the predictor of interest, where positive values indicate that the predictor of interest improves the model fit. * $P < .05$; ** $P < .01$; *** $P < .001$; **** $P < .0001$.

findings from similarly highly disadvantaged samples that described attenuated relationships of proximal factors (eg, negative parenting on child executive function,⁵¹ thrive factor [a composite including supportive parenting], and environmental stimulation-supporting cognition and language⁶) in greatly disadvantaged settings⁵¹ and highlight the importance of baseline levels of environmental resources for developmental trajectories.

Although brain volumes were related to cognitive outcomes at year 2, we did not find evidence that neonatal brain measures related to language scores or mediated associations between PSD and cognition or language. The absence of an

association between neonatal structural volumes and language at age 2 mirrors the null results of Spann et al⁷ in a smaller sample in the same age range. It is possible that global brain measures used in the current study (eg, total cortical GM) were not specific enough to detect these associations, and analyses of particular brain regions (ie, prefrontal or perisylvian cortex⁵²) may be more sensitive. Alternatively, investigations of different aspects of brain development, such as brain function, may be necessary to detect these associations.

This study had several strengths, including a large sample of 202 mother-child dyads who were oversampled for exposure to disadvantage, a multidimensional conceptualization of PSD,

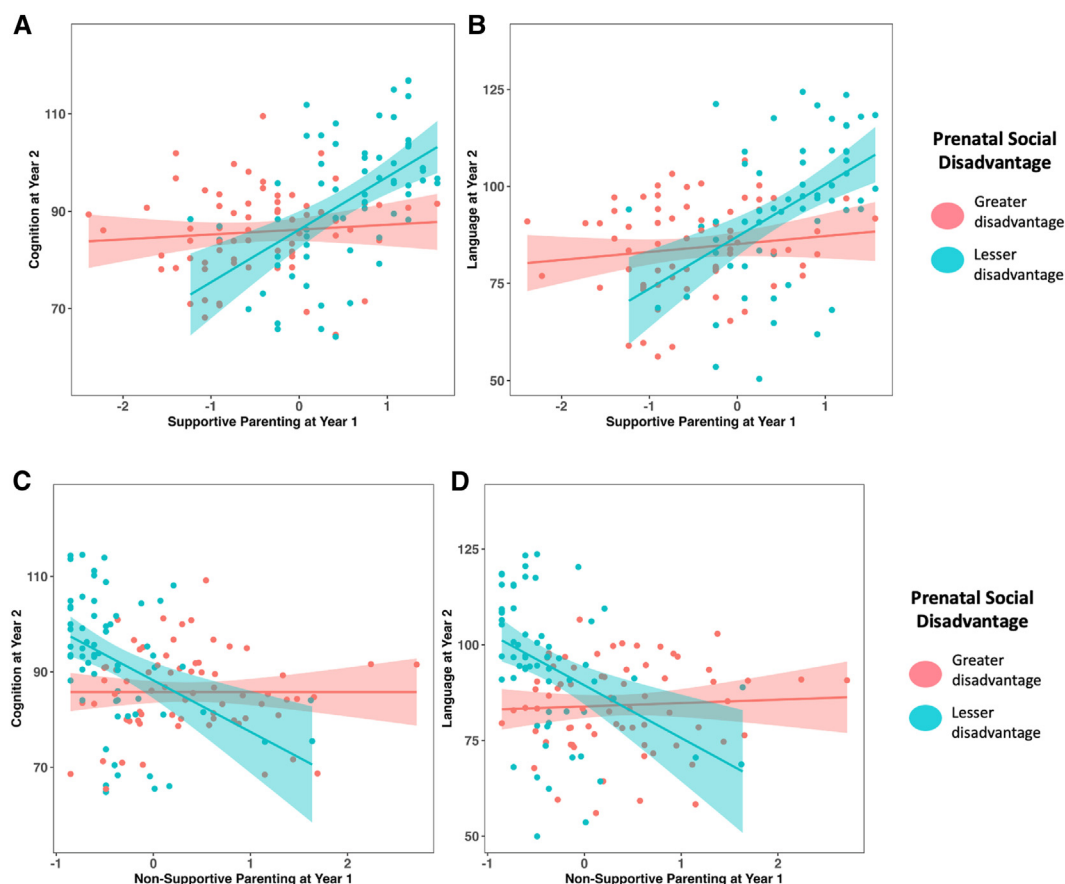


Figure 5. Prenatal social disadvantage (PSD) moderates the association between parenting behaviors and cognition and language scores. Interaction of PSD with supportive parenting behaviors to predict year 2 **(A)** cognition ($q < 0.000001$) and **(B)** language ($q < 0.000001$). Interaction of PSD with nonsupportive parenting behaviors to predict year 2 **(C)** cognition ($q < 0.00001$) and **(D)** language ($q < 0.0001$). PSD is split at median for visualization, however, all analyses used continuous PSD. Model: *gam(outcome ~ te(Prenatal_social_disadvantage, parenting) + Gestational_age + child_sex + age, method = "REML", fx = T)*.

investigations of nonlinear associations, and a longitudinal design to test mediation. However, there are also limitations. First, this study is correlational. As such, it is important to replicate these findings with future causal study designs to confirm

whether supportive parenting and/or poverty reduction interventions would be effective for supporting cognitive and language development. Moreover, like other data-driven models, GAMs may have limited generalizability to values that were

Table II. Parenting mediations in greater disadvantage vs lesser disadvantage

Group	Mediating variable	Outcome	Path A β	Path B β	Direct effect β	Indirect effect	95% CI indirect effect	
							Lower	Upper
Greater disadvantage (n = 71)	Supportive parenting behaviors	Cognition	-0.25	0.09	0.21	-0.02	-0.18	0.07
	Supportive parenting behaviors	Language	-0.25	0.12	0.27	-0.03	-0.19	0.09
	Nonsupportive parenting behaviors	Cognition	0.21	-0.02	0.19	-0.002	-0.08	0.10
	Nonsupportive parenting behaviors	Language	0.21	0.02	0.23	0.004	-0.67	0.08
Lesser disadvantage (n = 64)	Supportive parenting behaviors	Cognition	-.49***	.69***	-0.08	-0.34***	-0.54	-0.17
	Supportive parenting behaviors	Language	-.49***	.54**	-0.31	-0.26**	-0.49	-0.08
	Nonsupportive parenting behaviors	Cognition	.48***	-.49*	-0.18	-0.24*	-0.46	-0.03
	Nonsupportive parenting behaviors	Language	.48***	-.36	-0.4*	-0.17*	-0.36	-0.01

Standardized effect estimates (β) for path A, path B, direct and indirect effects. Path A: effect of the independent variable (prenatal social disadvantage) on the mediator variable. Path B: unique effect of the mediator variable on the dependent variable (outcome). Direct effect indicates the unique effect of independent variable (prenatal social disadvantage) on the dependent variable, while controlling for the mediating variable. The indirect effect is the mediation effect. Covariates in the model are sex, age at assessment, and gestational age at birth. Bolded values indicate a significant mediation effect. * $P < .05$, ** $P < .01$, *** $P < .001$, **** $P < .0001$.

not present in the original sample. Thus, other samples with different distributions of PSD may observe a different threshold above which parenting no longer influences outcomes. Second, PSD was correlated highly with disadvantage at year 1, so we could not investigate differential contributions of ongoing disadvantage. Third, consistent with previous studies that find that multiple aspects of parenting behaviors tend to load onto 2 broad dimensions,^{51,53,54} namely, supportive and nonsupportive, we examined parenting behaviors with 2 broad constructs. As a next step, future studies might tease apart associations between child outcomes and specific subcomponents of these composites (eg, sensitivity or positive regard) or other influential parenting behaviors not examined here (eg, parental structure or consistency). Moreover, parenting practices, and the perception of them by peers, assessors, and clinicians are sensitive to cultural beliefs and norms. Because a large proportion of parenting research has been conducted with white participants,¹⁵ the role and efficacy of parenting practices in different cultures has been less well-characterized. As such, studies such as this one rely on valuations of parenting behaviors that may not be generalizable across cultures. Future research could greatly benefit from creating and using culturally inclusive parenting assessments. Further, all the study's Parent Child Interaction video coders were white, although 59% of participants were Black. Although coders maintained high inter-rater reliability, future studies would benefit from assessment teams that match the demographic makeup of participants, which could foster cultural inclusivity in assessment practices. Finally, although sensitivity analyses found that PSD continued to predict cognition and language abilities over and above maternal cognitive ability, our measure of maternal cognition and PSD were highly correlated; thus, we cannot rule out potential genetic confounding. Future studies might use polygenic risk scores, twin studies, or adoption studies to further tease apart environmental vs genetic contributions.⁵⁵

Collectively, these findings suggest that prenatal disadvantage exerts meaningful effects on key developmental outcomes in early life. Parenting behaviors constitute one mechanism through which this occurs; however, this mechanism may only be impactful in lesser levels of disadvantage, suggesting that there may be a critical threshold beyond which mediating and moderating factors may become less effective. Thus, there is a need to address disadvantage as a primary prevention, bearing in mind that disadvantage encompasses multiple domains of social and material resources. ■

CRedit authorship contribution statement

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Declaration of Competing Interest

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Data Statement

Data sharing statement available at www.jpeds.com.

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